Challenging Attacker

Four attackers, five measurements, and the word that ranks them

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The epigraph I put on the Adversary branch is the whole argument in one line.

> *"If you know the enemy and know yourself, you need not fear the result of a hundred battles."*
>
> website, Social World Modeling, the Adversary entry, quoting The Art of War

Knowing the enemy is the expensive half. Red teams are scarce, hand-built attack scripts are slow to write and subjective to evaluate, and the automated adversaries that replace them are optimal, deterministic, and therefore learnable. A defender who has only ever trained against a learnable attacker is a defender who has trained against the wrong thing. That is what my index question asks.

> *"What makes human attackers uniquely challenging? How to best prepare defenders accordingly?"*
>
> website, Adversary, the Challenging Attacker entry

This file answers the first half by fixing four attackers and five measurements, and the second half by saying which of the four belongs in a training curriculum. Challenging is not a feeling here. It is a ranking over defenders, and it is defined below.

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1 What challenging means

An attacker is more challenging than another when a defender who has repeated experience against both ends up worse off against the first one. Three properties have to hold before the word is allowed.

1. **It is a claim about the defender, not the attacker.** Raw attacker strength is not the quantity. An attacker that wins on the first episode and is shut out by the two-thousandth is easy, whatever its opening score.
2. **It is measured after learning, not before.** The comparison is between the first five hundred episodes and the last five hundred. Every ranking in this file reverses somewhere in that interval, and reporting either end alone gets the ordering backwards.
3. **It has to survive the substitution of a human for the model.** A simulated defender that finds an attacker hard is a prediction. It becomes a finding when human participants, facing the same three attackers in the same task, reproduce the ordering.

The finding stated in my own paper title is that cognitive attackers are more challenging for defenders than strategic attackers. All three properties hold for it: the quantity is defender loss, it is read at the end of training, and human participants confirm the simulated ordering.

2 Active

2.1 The four attackers

B-line

> *"A fixed route."*
>
> website, Challenging Attacker

> *"Cheap to detect if you know the route. Almost impossible to detect from volume, because there isn't any."*
>
> website, Challenging Attacker

Definition: An attacker that assumes prior knowledge of the network topology and walks a prepared path to the operational server, User0 to User1 to Enterprise1 to Enterprise2 to Op_Server0, with a fallback table telling it where to resume when a step fails.

Distinguishes it from: Meander, which has no path. The separating property is prior knowledge of where the target sits, not aggression and not speed. Both attackers want the same server.

Operational test: Run the attacker twice from the same initial state with the defender asleep. If the two action sequences are identical, it is B-line.

In my project: The best known strategy in this scenario and the reference point for every reward figure. Against a passive defender it takes a mean reward of 112.8 with zero variance, which is the ceiling.

B-line on the website, Beeline in the papers

The page spells it *B-line*, with a hyphen, and gives it a section of its own. Both papers spell it *Beeline* and subscript it $Beeline_{Red}$, and the CAGE challenge that shipped it spells it Beeline as one word. The hyphenated form is mine and appears nowhere else. It is the better spelling, since the thing being named is a straight line rather than a bee, but a reader who greps the code for the page's spelling finds nothing. Fix one direction or the other.

Meander

> *"Breadth first. It scans every subnet it can see, discovers services on every address it has learned, exploits what it can, escalates where it can, and only then arrives at the same server. It leaves marks across the whole network on its way."*
>
> website, Challenging Attacker

> *"Loud, slow, and much harder to predict, because what it does next depends on what it happened to find."*
>
> website, Challenging Attacker

Definition: An attacker that assumes no knowledge of the network structure, picks its next target at random from what it has discovered, and takes privileged access on every host in a subnet before moving to the next one.

Distinguishes it from: B-line on prior knowledge, and from the cognitive attacker on learning. Meander is stochastic and never improves. Restarting it resets nothing, because it carries nothing between episodes.

Operational test: Compare episode one with episode two thousand against the same defender. If the behavior distribution has not shifted, the attacker is not learning, and stochastic is the most it can be.

In my project: The weakest of the three against every defender I have run. Against a cognitive defender it finishes the last five hundred episodes at zero reward and no impact at all, and it takes on average ten more steps than the cognitive attacker to hit each milestone.

Hard to predict against hard to defend

The page says Meander is much harder to predict. Both papers say Meander is the easiest of the three to shut down. Neither statement is wrong and the pair is the whole point of the branch, but on the page they sit under a heading that says Challenging Attacker, so a reader arrives at the wrong conclusion.

Unpredictability is a property of the action sequence. Difficulty is a property of the defender's loss. Meander is high on the first and lowest on the second, because randomness without adaptation is noise, and a defender learns to ignore noise faster than it learns to counter a plan. The attacker that is hard to defend against is the one that changes in response to what the defender did, which is neither of the two attackers described on this page.

cognitive attacker

Definition: An attacker whose action selection is a cognitive model of experiential choice rather than a script or an optimizer. In this line it is an instance-based learning agent: it stores each state, action, and outcome as an instance, retrieves past instances by activation, blends their outcomes into an expected value for each option, and takes the option with the highest blended value.

Distinguishes it from: A reinforcement learning attacker, which is the neighbor that gets confused with it most often. Both learn from interaction. A reinforcement learning algorithm is built to solve the problem optimally; a cognitive model is built to reproduce how a person solves it, which means it carries decay, retrieval noise, and recency as constraints rather than as defects. The two can reach the same policy and are not the same object.

Operational test: Remove the memory decay and the activation noise. If the agent still behaves the same, nothing cognitive was doing work and you have a value-based learner with extra vocabulary.

In my project: IBL_{Red}, the model I proposed. Trained against a passive defender for two thousand episodes it reaches a mean reward of 104.64 against B-line's 112.8, and by the end of training fifty five percent of the runs score above B-line. It learns that policy with no strategy encoded anywhere in it.

strategic attacker

Definition: An attacker whose policy is fixed in advance, whether it is deterministic like B-line or stochastic like Meander. Fixed means it does not condition on the defender's behavior and does not change across episodes.

Distinguishes it from: The cognitive attacker, on adaptivity alone. Stochastic is not adaptive. Both B-line and Meander are strategic in this sense, which is the point of the term: it groups the two attackers the website page describes and separates them from the one it does not.

Operational test: Freeze the defender's policy halfway through training. If the attacker's behavior distribution is unchanged by that, it was never conditioning on the defender, so it is strategic.

In my project: The comparison class in the title. Two members, and the finding holds against both of them separately.

> **Deterministic in 2023, strategic in 2025**
>
> The HICSS version of this work is titled *Human-Like Attackers are More Challenging for Defenders than Deterministic Attackers*. The journal version is titled *Cognitive Attackers are More Challenging for Defenders than Strategic Attackers*. Same finding, two vocabularies, and the substitution is not cosmetic.
>
> The earlier paper ran two attackers and the comparison class had one member, B-line, which really is deterministic. The later paper adds Meander, which is not deterministic, so the comparison class had to be renamed to something that covers both. *Strategic* is that name and it means fixed in advance. Anyone reading the two titles side by side will assume strategic is a synonym for deterministic. It is the wider word, and the wider claim is the one worth having, because the easy reply to the earlier paper is that stochasticity alone would have closed the gap. Meander is the control that says it does not.
>
> The same pass renamed the attacker: human-like became cognitive.

2.2 The defenders

passive defender

Definition: A defender that only monitors and never acts. It models the case where a stealthy attacker is present and undetected, and it exists to give a ceiling for each attacker rather than to defend anything.

Distinguishes it from: A defender that has learned to do nothing. This one was never able to act.

Operational test: Does its action distribution include any host-directed command. If not, it is passive.

In my project: $Sleepy_{Blue}$. Every attacker's training run and every reported ceiling is against it.

cognitive defender

Definition: The instance-based defender, built on the same theory as the cognitive attacker, whose state slots hold what a human defender would see: observed activity and known compromised status for each host, plus a step counter.

Distinguishes it from: The attacker model, which sees the network as an attack graph of access levels. The two agents share a theory and share no state representation, because they are not looking at the same thing.

Operational test: Show it an observation a human defender could not have had, such as the attacker's true privilege level on an unanalyzed host. If it consumes that, it is not this model.

In my project: IBL_{Blue}, from the HFES paper, reused here without modification so that the attacker comparison is the only thing that changes.

efficient defender

Definition: A human participant whose attacker held a reward below the mean for its condition. The split is per attacker type, so an efficient defender is efficient relative to others facing the same adversary.

Distinguishes it from: Expertise. The split is by outcome, not by background, and the sample was not selected for security experience.

Operational test: Compute the mean attacker reward within the condition, then sort. Below the mean is efficient.

In my project: The split that produced the sharpest result in the human experiment. Against the cognitive attacker, the efficient defenders are the group that suffers, and holding the line against it required a higher rate of active actions than either strategic attacker did.

2.3 The five measurements

Ranking attackers needs quantities that do not all say the same thing. These five were chosen so that an attacker can be ahead on one and behind on another, which is what happens.

reward

Definition: Cumulative attacker reward over the twenty five steps of an episode. Administrator access on a host pays 0.1, on a server 1.0, and a successful Impact on Op_Server0 pays 10.0 per step. Everything else pays nothing.

Distinguishes it from: Defender loss, which is its negative. The game is zero sum, so one series is enough and the defender panel is redundant.

Operational test: Sum the per-step payoffs at episode end.

In my project: The headline series. Against the cognitive defender, B-line opens at 112.8 and closes at 5.15; the cognitive attacker opens at 80.34 and closes at 54.60. The crossing is the finding.

impact duration

Definition: The number of steps per episode on which the attacker holds and impacts the operational server, counted after the first impact lands.

Distinguishes it from: Reward, which a patient attacker can accumulate from enterprise servers without ever reaching the target. Impact duration ignores everything except the objective.

Operational test: Count successive impact actions that succeed.

In my project: The persistence measure. In the last five hundred episodes against the cognitive defender it reads 3.25 for the cognitive attacker, 0.18 for B-line, and zero for Meander.

progress

Definition: The number of steps taken to first reach the enterprise subnet and, separately, the operational subnet. Two milestones, because the topology forces the attacker through the enterprise layer.

Distinguishes it from: Reward, which is cumulative and forgiving. Progress is a latency and is the quantity a defender actually manipulates: a defender wins by adding steps, not by subtracting points.

Operational test: Record the step index of first access to each subnet.

In my project: The measurement that shows the defense working. By the end of training the defender adds fifteen steps to B-line's route into the enterprise subnet and ten to the operational one, while the cognitive attacker's latencies barely move.

action frequency

Definition: The distribution over commands at each step of an episode, compared between early and late training.

Distinguishes it from: A performance metric. It explains a score rather than being one, and it is where a stalled attacker becomes visible.

Operational test: Plot the per-step command proportions for the first and last blocks. A distribution that has collapsed onto two commands is an attacker in a loop.

In my project: Late in training, both strategic attackers are caught in a loop of ExploitRemoteService and PrivilegeEscalate with the impact action gone. The cognitive attacker holds its distribution steady.

option space

Definition: The number of legal defender choices, each a command paired with a target host, available at a given step.

Distinguishes it from: The action space, which is a property of the game and is constant. The option space is a property of the situation the attacker has created, and the defender shrinks it by keeping hosts clean.

Operational test: Count valid command and host pairs at each step, averaged over the episode.

In my project: The load measure, and the one I would keep if I could only keep one. A defender that is winning is a defender whose option space shrinks with practice. Against both strategic attackers it shrinks. Against the cognitive attacker it does not move, so the defender is still choosing from a full menu at episode two thousand.

2.4 The finding, stated once

Three experiments, each answering one question, in the order they have to be asked.

1. **Can a cognitive model attack at all?** Against a passive defender the cognitive attacker starts near Meander and finishes near B-line, at 104.64 against 112.8, having learned the route from experience with nothing encoded. So it is a usable emulator, not a weak one.
2. **Is it harder for a learning defender?** Against the cognitive defender the ordering inverts within two thousand episodes. The defender drives both strategic attackers to near zero loss and only halves the cognitive attacker's take. Its action distribution against the strategic pair shifts toward early cheap removals; against the cognitive attacker it does not adjust at all.
3. **Do humans agree with the model?** One hundred eighty six participants on Mechanical Turk, one attacker each, seven episodes of twenty five steps after a practice pair. The human ordering matches the simulated ordering, including the crossing: B-line ahead in the first episode, behind by the last. The cognitive attacker does its best work against the defenders who are doing everything else right.

The answer to the second half of my index question follows from the third experiment. Prepare defenders against an adversary that adapts, because the skill that beats a fixed adversary is anticipation and it does not transfer. A cognitive attacker is cheap to run, needs no expert on call, and can be paired with a trainee's own history, which is what makes it a curriculum rather than a sparring partner.

3 Working

adversary emulation

Definition: The practice of standing up an attacker, human or automated, in order to test a defense rather than to breach it.

Distinguishes it from: Penetration testing, which aims at the network. Emulation aims at the defender, and its output is an assessment of defense rather than a list of vulnerabilities. The cross-domain entry for this term is in `dict_opponent_modeling.tex`.

instance-based learning theory

Definition: A theory of decisions from experience in dynamic tasks. Choices are made by recognizing similar past situations, blending their outcomes into an expected value per alternative, and taking the maximum.

Distinguishes it from: Expected utility maximization, which computes from stated probabilities. Here nothing is stated; everything comes from what happened before and how easily it is recalled.

instance

Definition: One unit of memory, a triple of situation, action, and experienced outcome.

Distinguishes it from: A transition tuple in reinforcement learning, which is consumed to update parameters and then discarded. An instance is kept and retrieved.

activation

Definition: The ease of retrieving an instance, a function of how recently and how often it was observed, plus noise.

Distinguishes it from: A value estimate. Activation governs whether a memory is available, not whether the outcome in it was good.

blending

Definition: The expected value of an option, formed as the retrieval probability weighted average of the outcomes stored in the instances for that option.

Distinguishes it from: A max over remembered outcomes. Blending averages, so a single lucky episode does not dominate.

delayed feedback

Definition: The condition where the outcome of an action is not available at the step it was taken, so the agent has to assign the eventual result back across the steps that produced it.

Distinguishes it from: Sparse reward. The signal here is not missing, it is late, and the credit assignment mechanism is what converts one into the other.

Interactive Defense Game

Definition: The browser task in which human participants play the defender side of this scenario, with the same network, the same four commands, and the same twenty five step episodes as the simulation.

Distinguishes it from: The simulation code the model defenders run under. The game exists so that human and model defenders can be compared without a translation step between them.

the adapted CAGE network

Definition: The seven-host reduction of the CAGE challenge network used in this line: four hosts and three servers over three subnets, with the operational subnet reachable only through the enterprise layer and the attacker starting on User0.

Distinguishes it from: CAGE Challenge 2 proper, which is thirteen hosts. The sibling Biased Attacker page uses the thirteen-host version, so the two pages under Adversary describe different networks under one scenario name.

> **One scenario name, two networks**
>
> This page and `bias/biased-attacker.md` both say CAGE and both draw a three-subnet picture with an enterprise gateway and Op_Server0 as the prize. The reward magnitudes agree exactly: 0.1 for a user host, 1.0 for a server, 10.0 for impact. The host counts do not. This line runs seven hosts, because an episode has to be short enough for the defender to observe a whole attack; the biased attacker line runs the full thirteen.
>
> That difference changes the option space by roughly a factor of two, and the option space is one of my five measurements. Any sentence that compares a number from one page against a number from the other has to say which network it is on.

4 Peripheral

attack graph

Definition: A representation of a network's vulnerabilities and the sequences that chain them, read from the attacker's side.

Pointer: Sheyner et al. 2002. It is where the attacker state slots in this line come from: subnet detected or scanned, host detected, scanned, exploited at user, exploited at root, impacted.

advanced persistent threat

Definition: A well resourced attacker that researches the target before acting, holds access over long periods, and plans against a known topology.

Pointer: The behavioral profile B-line is meant to stand for. Prior knowledge of the layout is the property that carries over, not the funding.

script kiddie

Definition: A novice attacker working from pre-made tools who scans broadly and exploits what turns up rather than planning a route.

Pointer: The profile Meander is meant to stand for. It is the reason Meander is stochastic rather than merely suboptimal.

active and passive defense actions

Definition: Remove and Restore are active, Analyse and Monitor are passive.

Pointer: The split that explains the human result. Efficient defenders facing the cognitive attacker used active actions at a higher rate than efficient defenders facing either strategic attacker, and human participants overall lean passive.

5 What the project holds

The code for this line sits in `Code/mentalworld-cyber-network-attackers`, a merge repository whose stated convention is to keep the two source trees separate and put comparison work in `integration/`. The tree for this lexicon is `sources/cyborg-ibl`, from the remote `DDM-Lab/CybORG-IBL`. The other tree belongs to the Biased Attacker line and its lexicon is `mwm_biased_attacker.tex`.

What is in it, by the file that holds it:

- `Cyborg.py` and `CyborgEnv.ipynb`, the simulation environment and its notebook.
- `IBLAttacker.py` and `IBLDefender.py`, the two cognitive agents.
- `HeuristicAgents.py`, which is where B-line, Meander, and the passive defender live. All three strategic agents are in one file because none of them learns.
- `myibl.py`, a modified instance-based learning implementation that adds delayed feedback. The stock library assumes the outcome arrives with the action, and in a twenty five step episode it does not.
- `Train.py` and `train_commands.sh`, the runs.
- `IBLBlue/`, four pickled defenders at 500, 1000, 1500, and 2000 episodes. Those are the checkpoints behind every early-against-late comparison in this file, and they are the reason the comparison can be rerun rather than re-trained.

The modified library and the four checkpoints are the two pieces worth naming to anyone who wants to reproduce this. Everything else is standard.

6 Sources

- Every quoted sentence above is from the website, either `opponent-agent-modeling/bias/adversary.md`, its parent `bias/README.md`, or the branch `README.md`. The locator on each one says which.
- Du, Y., Prébot, B., Malloy, T., and Gonzalez, C. A Cyber-War Between Bots: Cognitive Attackers are More Challenging for Defenders than Strategic Attackers. *ACM Transactions on Social Computing* 8(3-4), article 7, 2025. `doi:10.1145/3712672`. Read in full. Every number in the Active tier is from it.
- Du, Y., Prébot, B., Xi, X., and Gonzalez, C. Towards Autonomous Cyber Defense: Predictions from a Cognitive Model. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 66(1), 2022. `doi:10.1177/1071181322661504`. The cognitive defender reused here is from this paper.
- Du, Y., Prébot, B., Xi, X., and Gonzalez, C. A Cyber-War Between Bots: Human-Like Attackers are More Challenging for Defenders than Deterministic Attackers. *HICSS* 56, 2023, pages 856 to 865. The predecessor, read in full for the earlier vocabulary and for the clash on the two titles.
- Prébot, B., Du, Y., and Gonzalez, C. Learning about simulated adversaries from human defenders using interactive cyber-defense games. *Journal of Cybersecurity* 9(1), tyad022, 2023. The seven-host count and the human-defender material are from this one.

- The three tiers, the four fields, and the rule that a term whose two uses disagree belongs inside its own entry are from `_side/_lexicons/lexicon_method.tex`.